

National Science Foundation CyberCorps SFS Fellow

Andrew Strachan: Cybersecurity Systems Engineer & Secure Systems Architect

M.S. Cybersecurity Candidate (GPA 3.75) · University of Alabama at Birmingham

Dedicated to protecting national security and critical digital infrastructure. Bridging zero-trust systems engineering, on-device artificial intelligence, and pharmaceutical cGMP regulatory compliance with award-winning technical education leadership.



Interactive Resume & Technical Profile

Candidate credentials evaluated through interactive technology. Filter competencies across technical domains, review chronological milestones, and inspect verified STAR accomplishment evidence.

Andrew Strachan

CyberCorps: Scholarship for Service (SFS) Fellow · Information Security Analyst

Location: Birmingham, Alabama · Academic Contact: strachan@uab.edu · Pre-Vetted for Federal Security Clearances (Tier 5 SF-86 Track)

Academic Record & Degree Verification

Master of Science University of Alabama at Birmingham M.S. in Cybersecurity · GPA: 3.75 / 4.0 Dates: Aug 2024 – Expected Dec 2027 · NSF CyberCorps SFS Fellow. Key coursework: CS 623 Network Security, CS 646 Blockchain & Cryptography, CS 636 Threat Modeling, CJ 502 Forensics.	State Licensure University of Montevallo Teaching Field (PCTF): Business & Finance · GPA: 3.75 / 4.0 Dates: Aug 2024 – June 2025 · ALSDE Certified CTE Educator. Focus: Managerial finance, enterprise risk frameworks, curriculum engineering, and Alabama TAPE standards.	Bachelor of Science Mississippi College B.S. ACS Biochemistry Honors · GPA: 3.5 / 4.0 Dates: Aug 2011 – May 2016 · American Chemical Society Certified. Honors: Senior Class Senator, Symphonic Band Principal Horn, Phi Mu Alpha co-founder, pre-medical rigor.
Medical Training University of Mississippi School of Medicine M.D. Coursework & Clerkships (4 Years Completed) Dates: Aug 2016 – Dec 2020 · Completed in Good Standing. Honors: Human Anatomy Prosector, P.A.L.S. President, Virginia Covington Award; HIPAA security protocols.		

Filterable Technical Competency Matrix

▸ STRIDE Threat Modeling	▸ Zero-Trust Network Architecture	▸ Wireshark & Packet Inspection
▸ Nmap Port & Vulnerability Scanning	▸ SHA-256 Block Hashing Primitives	▸ NIST SP 800-53 / CSF 2.0 Compliance
▸ Digital Forensics & Artifact Extraction	▸ On-Device Local RAG Architecture	▸ Apple MLX & Metal Performance Shaders
▸ Quantized Gemma 4 / Ollama Execution	▸ SwiftData & Local SQLite Vector Search	▸ Model Context Protocol (MCP) Sandboxing
▸ AWS S3, CloudFront, Lambda, DynamoDB	▸ Firebase Hosting, Firestore, Functions	▸ Expiring HMAC Token Verification
▸ Docker Container Security & Isolation	▸ Swift & SwiftUI (Native macOS/iOS)	▸ Python (Systems, Data Analytics, Cryptography)
▸ Java 21 (Secure Sockets & Agent Gates)	▸ TypeScript & Modern ESNext	▸ C & Assembly Systems Fundamentals
▸ Microsoft Certified Educator (MCE)	▸ FBLA Chapter Advising & Coaching	▸ DECA State Competition Mentorship
▸ TAPE Standards Program Administration	▸ BBC micro:bit Physical Computing	

Chronological Professional Work History



Teaching Assistant — Python Coding Summer Camp

June 2026

University of Alabama at Birmingham (Department of Computer Science)

- Instructed middle school students in fundamental Python programming concepts, data structures, and algorithmic logic.
- Guided students to program BBC micro:bit microcontrollers to interface with sensors, LED matrices, and external robotics.
- Mentored students in writing custom Python flight trajectory scripts for aerial drones, demonstrating physical computing and hardware-software integration.



Career & Technical Education (CTE) Teacher — Business, Marketing & Finance

August 2024 – June 2025

Shades Valley High School (Jefferson County Board of Education - JEFCEOED)

- Instructed Business Finance, Customer Service, and Sales in the school's premier National Academy of Finance (NAOF) program.
- Served as official Chapter Adviser for the state's largest Future Business Leaders of America (FBLA) chapter.
- Head Coach for the Shades Valley High School Esports Team; organized fundraisers and secured corporate sponsorships.
- Maintained 100% compliance with Alabama State Department of Education TAPE instructional standards.
- Awarded the **JEFCEOED Technology Torchbearer Award (2025)** for innovative classroom technology integration.



Career & Technical Education (CTE) Teacher — Business, Marketing & Finance

September 2023 – June 2024

Corner High School (Jefferson County Board of Education - JEFCEOED)

- Founded, chartered, and advised a new DECA chapter, introducing competitive business leadership opportunities to the student body.
- Mentored and coached students in marketing and business finance simulations, culminating in **1st-place and 2nd-place state finishes** at the 2024 Alabama DECA Career Development Conference.
- Delivered experiential curriculum in financial software, personal finance, and entrepreneurship.
- Honored with the 2024 Alabama DECA New Chapter Advisor Recognition.



Operational Director

January 2023 – May 2023

MidSouth Extracts LLC · Tupelo, MS

- Directed operations for the establishment of a state-licensed, cGMP-regulated botanical and medical processing facility.
- Authored a comprehensive, audit-ready library of Standard Operating Procedures (SOPs) spanning hazardous extraction, QA/QC testing, and employee hygiene.
- Embedded with PhD analytical chemists and engineers to execute detailed process hazard and workflow analyses.
- Secured complex state and municipal permits with 100% first-pass regulatory audit compliance.

Sales Development Specialist & Team Lead

April 2021 – November 2022

SelectQuote Insurance Services · Clinton, MS

- Developed, evaluated, and led onboarding and training curricula for incoming sales cohorts across life and health divisions.
- Partnered with compliance leadership to engineer micro-training modules reducing customer complaints and raising client retention.
- Awarded the **Top Sales Award (2022)** and **SelectQuote Team Leader Award (2023)**.

Founder & Lead Technologist

2022 – Present

Maqkrs Consulting · Birmingham, AL & Clinton, MS

- Provided strategic operational consulting on FDA, ISO, and cGMP compliance frameworks for regional startups.
- Engineered custom digital operations: cloud hosting, web platforms, marketing funnels, and licensing pipelines.
- Mentored students in work-based learning across vulnerability scanning and zero-trust configuration checklists.

Measurable Accomplishments (STAR Methodology)

STAR-01

JEFCOED Honors

Technology Torchbearer Award (2025)

Situation: Shades Valley HS required modern tech integration across CTE classrooms.

Task: Deploy experiential business software curriculum and prepare students for FBLA competitions.

Action: Introduced interactive spreadsheet simulation models and mentored the state's largest FBLA chapter.

Result: Honored by the Jefferson County Board of Education with the 2025 Technology Torchbearer Award.

STAR-02

State Competition

State Championship DECA Coaching

Situation: Corner HS lacked competitive business leadership pathways.

Task: Charter a new DECA chapter and train students for statewide competition.

Action: Coached weekly business judging simulations, analyzing market financial case studies.

Result: Students earned 1st-place and 2nd-place state trophies at the 2024 Alabama DECA Career Development Conference.

STAR-03

Regulatory QA

cGMP Facility Buildout & SOP Library

Situation: MidSouth Extracts required full FDA and state health regulatory licensure.

Task: Author audit-ready Standard Operating Procedure library prior to inspection.

Action: Synthesized complex extraction protocols into 30+ SOPs with PhD analytical chemists.

Result: Passed all state audits on the first pass with 100% compliance and zero citations.

STAR-04

3D Systems

CS646 Sanctum 3D Memory Palace

Situation: Graduate students struggled to visualize distributed ledger consensus mechanisms.

Task: Build an interactive 3D WebGL simulation demonstrating cryptographic primitives.

Action: Engineered Sanctum in Godot 4 WebGL, integrating real-time SHA-256 block hashing.

Result: Live AWS CloudFront deployment with 720p 30fps web previews and permanent course adoption.

STAR-05

Edge Privacy

MaqkrsTutor2 Local AI Study Assistant

Situation: Commercial AI cloud tools leak institutional IP and violate student privacy.

Task: Architect a native macOS AI study tool with zero network egress.

Action: Engineered SwiftUI app with local quantized Gemma 4 and SQLite vector retrieval.

Result: Presented at UAB CS Spring Showcase (ACM & WiT) with sub-second local retrieval and 100% privacy.

STAR-06

Cloud Security

Secure File Sharing with Expiring HMAC Tokens

Situation: Static cloud file sharing links enable unauthorized data exfiltration.

Task: Build serverless file exchange with cryptographic signature expiration.

Action: Implemented AWS S3, DynamoDB, and Lambda generating 15-minute expiring HMAC tokens.

Result: Zero unauthorized access vulnerabilities and automated S3 lifecycle garbage collection.

Career Research & Future Outlook

Rigorous labor market investigation for the chosen career pathway: Information Security Analyst & Cybersecurity Systems Engineer, substantiated by official U.S. Bureau of Labor Statistics (BLS) data, federal compensation scales, and industry challenge analysis.

Target Career: Information Security Analyst & Secure Systems Architect

Standard Occupational Classification: BLS SOC 15-1212.00 | CyberCorps SFS Public Service Specialization

Information Security Analysts plan, engineer, implement, and monitor defensive security measures to protect computer networks, cloud architectures, and digital assets. Aligned with the National Initiative for Cybersecurity Education (NICE) Workforce Framework categories: **PR-CD-001 (Cyber Defense Analyst)** and **SP-ARC-002 (Security Architect)**.

Zero-Trust Network Access

STRIDE Threat Modeling

NIST SP 800-53 Controls

Vulnerability Remediation

Post-Quantum Cryptography

Official BLS Economic Research & Job Outlook (SOC 15-1212.00)

\$120,360

MEDIAN ANNUAL WAGE

\$57.87 / hour (BLS OOH
National Median)

32%

PROJECTED GROWTH
(2022–32)

Much faster than average (all
jobs: 3%)

\$182,370+

TOP DECILE WAGE (90TH
%)

\$87.68+ / hour (Principal
Systems Architect)

+53,200

PROJECTED NEW JOBS

~16,800 average annual job
openings

EARNING TIER / PERCENTILE	ANNUAL WAGE (USD)	HOURLY EQUIVALENT	PROFESSIONAL CONTEXT & TITLE
Lowest 10%	\$69,310	\$33.32 / hr	Junior Security Analyst / Associate SOC Analyst
25th Percentile	\$89,040	\$42.81 / hr	Cyber Defense Analyst / Network Security Specialist
50th Percentile (Median)	\$120,360	\$57.87 / hr	Information Security Analyst / Incident Responder
75th Percentile	\$152,430	\$73.28 / hr	Senior Security Systems Engineer / Threat Modeler
Highest 10% (Top Decile)	\$182,370+	\$87.68+ / hr	Principal Security Architect / Federal Technical Lead

Federal Public Service Pay Bands (General Schedule GS-9 to GS-14)

Funded by the National Science Foundation, CyberCorps SFS provides Direct Hire Authority (DHA) for federal service placements upon graduate degree completion at UAB.

GS-09 Entry Band

\$65,000 – \$84,000

Cybersecurity Graduate Intern / Junior Federal Analyst. Requires Master's candidate status; active SFS fellow pre-clearance vetting.

GS-11 / GS-12 Target

\$78,000 – \$122,000

Information Security Systems Engineer / Cyber Defense Specialist. SFS graduate placement with full Top Secret / SCI clearance.

GS-13 / GS-14 Senior

\$110,000 – \$168,000+

Senior Security Architect / Agency Technical Lead. Includes federal Cyber Incentive Pay bonuses up to 25% for high-demand technical roles.

Industry Obstacles & Engineering Mitigations

CRITICAL INDUSTRY CHALLENGE 01

Post-Quantum Cryptography Migration

Quantum computer scalability threatens RSA and ECC asymmetric encryption, enabling 'harvest now, decrypt later' adversary tactics against federal systems.

Mitigation:

Research into NIST-standardized lattice-based algorithms (ML-KEM, ML-DSA). Engineered in CS 646 Sanctum 3D cryptographic visualization engine.

CRITICAL INDUSTRY CHALLENGE 02

Agentic AI Weaponization & Automated Exploits

Adversaries deploy autonomous LLM agents that discover zero-day bugs, synthesize polymorphic malware, and evade traditional firewall rules.

Mitigation:

Architected 'Citadel' zero-trust gatekeeper (Java 21 / MCP) enforcing sandboxed tool authorization and privacy-first local models (MaqkrsTutor2).

CRITICAL INDUSTRY CHALLENGE 03

Critical Infrastructure Ransomware Defense

National SCADA, healthcare, and educational IT systems face sophisticated ransomware cartels targeting unsegmented legacy networks.

Mitigation:

Enforcing defense-in-depth, STRIDE threat models, and pharmaceutical cGMP-derived audit SOPs to guarantee resilient operational continuity.

Sample Materials: Career-Related Education

Academic coursework, research projects, and business education applied to real systems, accompanied by detailed analysis of their direct impact on Andrew's future cybersecurity career.

UAB Cybersecurity Graduate Coursework

Department of Computer Science (CS 623, CS 646, CS 636, CS 532) & Criminal Justice (CJ 502)

Advanced graduate studies in Network Security, Blockchain & Cryptography, Computer Security Threat Modeling, Cloud Computing, and Digital Forensics.

Impact on Future Career:

Directly establishes the technical defense capabilities required for Information Security Analysts. Imparts hands-on mastery in packet inspection, TLS handshake negotiation, cryptographic primitives, and forensically sound evidence chains.

STRIDE Threat Modeling Portfolio (CS 636)

Enterprise Systems Security Architecture & Attack Boundary Deconstruction

Authored structured Level 0 and Level 1 Data Flow Diagrams (DFDs) analyzing attack vectors across BlazerNET enterprise network and IoT camera endpoints.

Impact on Future Career:

Instills proactive security engineering instincts to anticipate adversary attack vectors before software deployment. Equips Andrew to evaluate enterprise software against OWASP Top 10, NIST SP 800-53 controls, and federal zero-trust mandates.

University of Montevallo CTE Business Education

Provisional Certificate in a Teaching Field (PCTF) — Business, Marketing & Finance

Completed graduate education in managerial accounting, economic forecasting, curriculum engineering, and Alabama TAPE governance standards (GPA 3.75).

Impact on Future Career:

Provides indispensable executive communication, financial modeling, and instructional leadership skills. In federal cybersecurity, analysts must translate complex technical cyber risks to non-technical executives and train personnel on cyber hygiene.

Mississippi College ACS Biochemistry & UMMC Medicine

Honors Baccalaureate & Clinical Sciences Foundation (Master Prosector)

Four years of preclinical medical coursework, clinical clerkships, master cadaveric dissection, and ACS-certified biochemistry laboratory protocols.

Impact on Future Career:

Instills scientific empirical hypothesis testing and zero-defect attention to detail. Operating in sterile surgical suites and cGMP extraction facilities directly mirrors the discipline required to enforce strict zero-trust postures and safeguard sensitive health data (HIPAA).

Sample Materials: Educational Enhancement

Structured career development planning, federal opportunity tracking, international humanitarian service, youth STEM mentoring, and tangible products developed during experiential learning.

51 Tracked Opportunities

51-Position Federal Opportunity Tracker

Documented in `accurateinternshiptracker.xlsx`: tracks positions across Dream Tier (White House OSTP, ONCD, USDS), National Labs (Sandia TITANS, Los Alamos CAI, Oak Ridge), and Defense (U.S. Army Space & Missile Defense Command, Redstone Arsenal, CISA).

Career Relevance:

Demonstrates structured, long-term career execution fulfilling the CyberCorps SFS public service service requirement across GS-9 through GS-12 bands.

14 Federal Drops

14 SFS Virtual Job Fair Applications

Submitted tailored applications to 14 premier federal intelligence, defense, and civilian agencies: U.S. Space Force, CISA, MARFORCYBER, DOJ CCIPS Cyber Crime Lab, CBP Cyber Threat Intelligence, and Sandia National Laboratories.

Career Relevance:

Direct engagement with federal recruiters securing summer graduate internships that convert into permanent federal civil service roles.

Global Leadership

International Humanitarian Service

Uganda health mission establishing 2 mobile clinics serving 1,000+ rural patients; U.S. Department of State Benjamin A. Gilman International Scholarship in Pokhara, Nepal conducting hospital clinical service and educational broadcasts for Reach the World.

Career Relevance:

Cultivates global situational awareness, high-stakes operational composure, and community stewardship foundational to public service leadership.

STEM Mentorship

Youth STEM Drone & Python Mentoring

Teaching Assistant for UAB Python Summer Camp (June 2026), instructing middle school students in algorithmic coding, BBC micro:bit microcontrollers, and autonomous aerial drone trajectory programming.

Career Relevance:

Validates hands-on hardware-software integration capabilities and commitment to inspiring the next generation in computer science.

Sample Materials: Special Skills & Credentials

Five curated special skills strictly correlated to the federal cybersecurity mission, anchored by an official Credly-verified Microsoft Certified Educator (MCE) digital credential and 30 verified LinkedIn Learning certifications.

Verified Industry Credential



Microsoft Certified Educator (MCE)

Issuer: Microsoft Corporation & Certiport · Issued May 16, 2025 · Verified via Credly

Validates technology literacy and pedagogical competencies needed to architect custom learning environments. Proves Andrew's ability to design enterprise cybersecurity awareness programs, conduct federal workforce upskilling, and translate technical security controls into operational standards.



Verify Credly Badge (Public URL)

(https://www.credly.com/badges/d4e5c326-c255-405c-b50e-0a369d6fc3a0/public_url)

Top 5 Specialized Skills Aligned to Federal Mission

1. Secure Systems & Zero-Trust Architecture

Credential: NSF CyberCorps SFS Fellowship & M.S. Cybersecurity (GPA 3.75)

Enforces OMB M-22-09 zero-trust directives through STRIDE threat modeling, packet inspection (Wireshark), port auditing (Nmap), and defense-in-depth principles.

2. Educational Technology & Instructional Design

Credential: Microsoft Certified Educator (Credly Verified) & Alabama PCTF

Architects technical security training programs, translates technical controls into standard operating procedures, and trains non-technical agency personnel.

3. On-Device AI & Privacy-Preserving Systems

Credential: Native Apple Silicon (MaqrsTutor2) & 30 Professional AI Certs

Prevents sensitive data egress by deploying local RAG architectures, quantized Gemma 4 models, and hardware-accelerated inference via Apple MLX.

4. Cloud Infrastructure & DevSecOps Engineering

Credential: CS 532 Cloud Systems & Multi-Cloud Production Deployments

Architects least-privilege AWS/GCP/Firebase environments with expiring HMAC token verification, Docker isolation, and automated CI security gates.

5. Regulatory Compliance & Quality Systems

Credential: MidSouth Extracts Operational Directorship & cGMP SOP Library

Bridges technical cybersecurity controls with legal and regulatory mandates, applying pharmaceutical cGMP discipline to federal GRC and digital forensics.

Directory of 30 Verified LinkedIn Learning Professional Certifications

All 30 certifications have been verified live (HTTP 200) matching recipient identity and course curricula across Agentic AI, Cloud Core Services, and AI Governance.

Developing Executive Presence (https://www.linkedin.com/learning/certificates/80c5af13023661b0f58775b2a44868b0b94ead17c194f21ee453d156896b27af74b588d752bd4b14620a0f9743af0727627c404a349162332ad4)	How to Make Innovative Thinking a Habit (https://www.linkedin.com/learning/certificates/58156896b27af74b588d752bd4b14620a0f9743af0727627c404a349162332ad4)	Tech Trends (https://www.linkedin.com/learning/certificates/58156896b27af74b588d752bd4b14620a0f9743af0727627c404a349162332ad4)
Fundamentals of Agentic AI (https://www.linkedin.com/learning/certificates/a540ca10a00b21b0f58775b2a44868b0b94ead17c194f21ee453d156896b27af74b588d752bd4b14620a0f9743af0727627c404a349162332ad4)	Generative AI with Multi-Agent LangChain (https://www.linkedin.com/learning/certificates/a540ca10a00b21b0f58775b2a44868b0b94ead17c194f21ee453d156896b27af74b588d752bd4b14620a0f9743af0727627c404a349162332ad4)	Building Agents with Vertex AI (https://www.linkedin.com/learning/certificates/a540ca10a00b21b0f58775b2a44868b0b94ead17c194f21ee453d156896b27af74b588d752bd4b14620a0f9743af0727627c404a349162332ad4)
How to Boost Productivity with AI (https://www.linkedin.com/learning/certificates/44d54d96d179a4e0b54701e725b1288c5047b6686e94259fca25734f954c4342089701f31616c1d3889ed709f09a2e036c4d572b082303e4)	Greg McKeown on Effortless Results (https://www.linkedin.com/learning/certificates/44d54d96d179a4e0b54701e725b1288c5047b6686e94259fca25734f954c4342089701f31616c1d3889ed709f09a2e036c4d572b082303e4)	Nano Tips for Using AI (Rachel Woods) (https://www.linkedin.com/learning/certificates/44d54d96d179a4e0b54701e725b1288c5047b6686e94259fca25734f954c4342089701f31616c1d3889ed709f09a2e036c4d572b082303e4)

▶ Power Platform Fundamentals (PL-900) (https://www.linkedin.com/learning/certificates/cf971c20e18b737b44e479209110e311a7247f984b9994a071e26e5422a22c59938d37189p270566864#583e11b4b0f529039j1a480d10d5783bf21259d)	▶ What Is Generative AI? (https://www.linkedin.com/learning/certificates/4b113e7297290b18b27f727a99599b5276e63a653757b34941467059270566864#583e11b4b0f529039j1a480d10d5783bf21259d)	▶ Generative AI vs. Traditional AI (https://www.linkedin.com/learning/certificates/4b113e7297290b18b27f727a99599b5276e63a653757b34941467059270566864#583e11b4b0f529039j1a480d10d5783bf21259d)
▶ Generative AI in Finance & Accounting (https://www.linkedin.com/learning/certificates/62c8b62d8af996b7c297290b18b27f727a99599b5276e63a653757b34941467059270566864#583e11b4b0f529039j1a480d10d5783bf21259d)	▶ Accounting Foundations (https://www.linkedin.com/learning/certificates/62c8b62d8af996b7c297290b18b27f727a99599b5276e63a653757b34941467059270566864#583e11b4b0f529039j1a480d10d5783bf21259d)	▶ Accounting Foundations: Bookkeeping (https://www.linkedin.com/learning/certificates/62c8b62d8af996b7c297290b18b27f727a99599b5276e63a653757b34941467059270566864#583e11b4b0f529039j1a480d10d5783bf21259d)
▶ Advanced AI Governance & Controls (https://www.linkedin.com/learning/certificates/91b82fcd090789b7c297290b18b27f727a99599b5276e63a653757b34941467059270566864#583e11b4b0f529039j1a480d10d5783bf21259d)	▶ Agentic AI Business Strategy (Charlene Li) (https://www.linkedin.com/learning/certificates/91b82fcd090789b7c297290b18b27f727a99599b5276e63a653757b34941467059270566864#583e11b4b0f529039j1a480d10d5783bf21259d)	▶ Copilot for Service (https://www.linkedin.com/learning/certificates/91b82fcd090789b7c297290b18b27f727a99599b5276e63a653757b34941467059270566864#583e11b4b0f529039j1a480d10d5783bf21259d)
▶ AI for React Developers (https://www.linkedin.com/learning/certificates/397b40ed680f5b7c297290b18b27f727a99599b5276e63a653757b34941467059270566864#583e11b4b0f529039j1a480d10d5783bf21259d)	▶ Knowledge Graphs for Generative AI (https://www.linkedin.com/learning/certificates/397b40ed680f5b7c297290b18b27f727a99599b5276e63a653757b34941467059270566864#583e11b4b0f529039j1a480d10d5783bf21259d)	▶ Google Cloud Core Services (https://www.linkedin.com/learning/certificates/397b40ed680f5b7c297290b18b27f727a99599b5276e63a653757b34941467059270566864#583e11b4b0f529039j1a480d10d5783bf21259d)
▶ Agentic AI for Developers (https://www.linkedin.com/learning/certificates/32bfa0c9bbe8c4b7c297290b18b27f727a99599b5276e63a653757b34941467059270566864#583e11b4b0f529039j1a480d10d5783bf21259d)	▶ Preparing Your Organization for AI Agents (https://www.linkedin.com/learning/certificates/32bfa0c9bbe8c4b7c297290b18b27f727a99599b5276e63a653757b34941467059270566864#583e11b4b0f529039j1a480d10d5783bf21259d)	▶ Upskilling Your Organization on AI (https://www.linkedin.com/learning/certificates/32bfa0c9bbe8c4b7c297290b18b27f727a99599b5276e63a653757b34941467059270566864#583e11b4b0f529039j1a480d10d5783bf21259d)
▶ Future of Workforce Learning (https://www.linkedin.com/learning/certificates/e727c26a36b402b5baf75a09410431b3a27f727a99599b5276e63a653757b34941467059270566864#583e11b4b0f529039j1a480d10d5783bf21259d)	▶ Generative AI in Learning & Development (https://www.linkedin.com/learning/certificates/e727c26a36b402b5baf75a09410431b3a27f727a99599b5276e63a653757b34941467059270566864#583e11b4b0f529039j1a480d10d5783bf21259d)	▶ Using Gen AI for Personalized Learning (https://www.linkedin.com/learning/certificates/e727c26a36b402b5baf75a09410431b3a27f727a99599b5276e63a653757b34941467059270566864#583e11b4b0f529039j1a480d10d5783bf21259d)
▶ Building AI Apps with Prompt Flow (https://www.linkedin.com/learning/certificates/638879edcc13d3b7c297290b18b27f727a99599b5276e63a653757b34941467059270566864#583e11b4b0f529039j1a480d10d5783bf21259d)	▶ Finding & Retaining High Potentials (https://www.linkedin.com/learning/certificates/638879edcc13d3b7c297290b18b27f727a99599b5276e63a653757b34941467059270566864#583e11b4b0f529039j1a480d10d5783bf21259d)	▶ Google Gemini for Business Analysis (https://www.linkedin.com/learning/certificates/638879edcc13d3b7c297290b18b27f727a99599b5276e63a653757b34941467059270566864#583e11b4b0f529039j1a480d10d5783bf21259d)

Interactive Media & 28+ Project Showcase

Explore Andrew's real technical body of work. Inspect embedded 10-second web video previews with dual MP4/WebM formats, followed by a verified live directory of 28+ cloud-hosted applications.

Curated Highlight Projects with Video Previews

CS646 Sanctum 3D Memory Palace

Gamified Cryptographic Distributed Ledger Environment

Godot 4

WebGL

AWS CloudFront

SHA-256

TypeScript

Interactive 3D exploration world featuring 5 distinct districts, procedural hover traversal, and real-time SHA-256 block hashing demonstrations.

MaqkrsTutor2 (Local AI Study Assistant)

Native Apple Silicon Local RAG Study Assistant

SwiftUI

SwiftData

Apple MLX

Quantized Gemma 4

Ollama

Native assistant indexing documents and code for semantic query retrieval with 0 network egress, guaranteeing 100% privacy and institutional IP security.

AdaptivePrep & AdaptiveHS Learning Portal

Full-Stack K-12 Homeschool Learning Management Platform

Next.js 15

Firebase Firestore

Cloud Run

Tailwind CSS

DemoAuth

Full-stack homeschool portal aligned with Georgia Standards of Excellence (GSE). Features multi-role dashboards for parents, students, and educators.

MaQkRs OS Command Center & Planner

Executive Dashboard & Multi-Agent Operations Console

React 19

TypeScript

Firebase

Tailwind CSS

Python

Unified operational command center providing real-time telemetry across multi-agent workflows, business financials, and project sprint schedules.

Directory of 28+ Verified Live Project Deployments

Every project link below is live-checked (HTTP 200) across AWS CloudFront, Firebase Hosting, Vercel, and Lovable platforms.

CS646 Sanctum 3D World (AWS)

(<https://d1puxvoeqwchkj.cloudfront.net/>)

AWS CloudFront · Godot WebGL

Primary 3D memory palace exploring cryptographic consensus.

CS646 Sanctum (Firebase Fallback)

(<https://cs646-sanctum-world.firebaseio.com/>)

Firebase Hosting · WebGL

Alternative high-availability node for the Sanctum 3D world.

CS646 Blockchain Study Hub

(<https://cs646-sanctum-astrachan.firebaseio.com/>)

Firebase Hosting · React

Permanent course portal for Dr. Yuliang Zheng's curriculum.

CS646 Midterm Review Portal

(<https://cs646-sanctum-astrachan--midterm-aogm2izm.web.app/#/>)

Firebase Web App · Single Page

Cryptographic lexicons, study guides, and flashcards.

Quantum Universe 3D Dashboard

(<https://quantum-universe-3d.web.app/>)

Firebase Web App · Three.js

Cyber-themed spatial operations console with live nodes.

MaQkRs OS Command Center

(<https://maqkrs-command-center.firebaseio.com/>)

Firebase Hosting · React 19

Multi-agent business operations and schedule dashboard.

AdaptivePrep K-12 Platform

(<https://adaptivehs-v2.firebaseio.com/>)

Cloud Run · Next.js 15

Homeschool portal aligned with Georgia GSE standards.

AdaptiveHS Demo Simulation

(<https://d3jeotfnsm148g.cloudfront.net/work/adaptiveh/>)

AWS CloudFront · Static Export

Static walkthrough simulation of homeschool role dashboards.

Project Atlas 3D Portfolio

(<https://d3jeotfnsm148g.cloudfront.net/>)

AWS CloudFront · Spatial Storytelling

Scroll-driven canvas architecture with interactive case studies.

Coastal Speech Co. (<https://coastal-speech-co-fl.firebaseio.com/>)

Firebase Hosting · React

Clinical practice web platform for Panama City, FL speech clinic.

Victorious Herbal Elements

(<https://aether-apothecary.vercel.app/>)

Vercel · Next.js Storefront

Client botanical product catalog and checkout workflow.

Networking Universe (CS 334/534)

(<https://networking-testy.firebaseio.com/>)

Firebase Hosting · 3D Visualizer

Interactive 3D network topology and packet routing simulation.

CS 623 Network Security Hub

(<https://networking-security-cs623-hub.firebaseio.com/>)

Firebase Hosting · Markdown Hub

Course resource platform for firewalls and network defense.

Switch MAC Address Learning

(<https://practice-questions-networking.firebaseio.com/>)

Firebase Hosting · Interactive Canvas

Interactive Layer 2 switch forwarding table simulator.

Networking Midterm Guide

(<https://networking-midterm.web.app/studyguide4.html>)

Firebase Web App · Deep Link

Intensive study guide with 33 question variant modules.

CS 334/534 Study Resources

(<https://networking-exam-60418.web.app/>)

Firebase Web App · Static Hub

Static reference platform organizing network exam guides.

CS 423/623/723 Practice Test

(<https://network-security-study.firebaseio.com/>)

Firebase Hosting · 200-Q Test

Network security examination engine testing defense-in-depth.

Network Security Demos (<https://ns-demos-34811.firebaseio.com/>)

Firebase Hosting · State Engine

Interactive firewall, proxy, and access control list (ACL) demos.

Cyber Risk Management (NIST)

(<https://cyberriskmanagement-testy.firebaseio.com/>)

Firebase Hosting · NIST Portal

UAB risk study hub mastering NIST SP 800-30 and CSF.

CS 203 OOP Study Hub (<https://cs203-java-studyguide.firebaseio.com/>)

Firebase Hosting · Code Snippets

Java OOP guide covering inheritance and encapsulation.

CS 203 Practice Question Bank

(<https://cs203-java-practice1.firebaseio.com/>)

Firebase Hosting · Question Bank

Interactive Java OOP drill examining object reference models.

CS 203 OOP Exam Simulator

(<https://cs203-exam-simulator.firebaseio.com/>)

Firebase Web App · Timed Simulator

Timed exam simulator grading complex Java programs.

CS 203 Code Output Analysis (https://cs-203-midterm.web.app/practice_test_labs.html)

Firebase Web App · Deep Path

Diagnostic tool testing student ability to trace Java code.

CS 203 Exam 1 Practice Test (https://cs-203-midterm.web.app/practice_test.html)

Firebase Web App · Practice Test

Practice test covering loops, class inheritance, and types.

Computer Security Companion

(<https://secure-connector-buddy.lovable.app/>)

Lovable App · AI Study Tool

Interactive AI companion for CS 436/636/736 computer security.

Cybersecurity Study Guide (<https://andy-sg-ns.firebaseio.com/>)

Firebase Hosting · Question Engine

Question variant generator covering security fundamentals.

Andrew Strachan GitHub

(<https://github.com/astrachan163>)

GitHub · Developer Profile

Open source repositories, systems scripts, and academic tools.

Microsoft Credly Badge

(https://www.credly.com/badges/d4e5c326-c255-405c-b50e-0a369d6fc3a0/public_url)

Credly Verification Platform

Official Microsoft Certified Educator digital badge verification.

Sources, Provenance & FBLA Rubric Audit

Full audit trail substantiating every factual claim, credential, labor statistic, and code artifact against professionally legitimate authorities (BLS, NIST, NSF, Microsoft Credly), accompanied by a self-evaluating FBLA 100-Point Rating Sheet.

FBLA 100-Point Rating Sheet — Evaluated at "Exceeds Expectations" (100 points / 100 pts)

FBLA EVALUATION CATEGORY	RATING LEVEL	SCORE	PORTFOLIO EVIDENCE & ARCHITECTURAL IMPLEMENTATION
1. Resume	Exceeds Expectations	10 / 10	Interactive technology filtering across 5 technical domains, chronological timeline, verified GPAs (UAB 3.75, Montevallo 3.75, MC 3.5), and STAR accomplishments.
2. Career Research	Exceeds Expectations	10 / 10	Official BLS SOC 15-1212.00 economic data (\$120,360 median, \$182,370+ top decile, 32% growth), federal GS-9 to GS-14 pay progression, and 3 critical industry obstacles with mitigations.
3. Career-Related Education	Exceeds Expectations	15 / 15	UAB graduate coursework (CS 623, 646, 636, CJ 502), STRIDE threat modeling, Montevallo CTE finance, and ACS biochemistry, with detailed analysis of impact on future cybersecurity career.
4. Special Skills & Endorsements	Exceeds Expectations	15 / 15	Top 5 specialized skills correlated to federal mission, anchored by Credly-verified Microsoft Certified Educator (MCE) badge embed and 30 verified LinkedIn Learning certifications.
5. Sources & Citations	Exceeds Expectations	10 / 10	Substantiated by legitimate authorities: Bureau of Labor Statistics (BLS), National Institute of Standards and Technology (NIST SP 800-53 / CSF), NSF CyberCorps SFS, and Credly. 100% HTTPS URLs.
6. Use of Portfolio in Presentation	Exceeds Expectations	10 / 10	Integrated Presenter HUD with official 7-minute countdown timer (420s), 60s warning alert, section-synchronized speaker notes, presentation clicker hotkeys, and PDF download.
7. Delivery & Flow	Exceeds Expectations	10 / 10	Flawless logical flow: Sticky Nav → Hero Identity → Interactive Resume → Career Research → Sample Materials → Media Showcase → Sources & Audit Ledger.
8. Educational Enhancement	Exceeds Expectations	10 / 10	51-position federal opportunity tracker, 14 SFS job fair applications, Uganda health outreach, Gilman Scholarship in Nepal, youth STEM mentoring, and 4 working software products.
9. Technical Polish & Accessibility	Exceeds Expectations	10 / 10	Cyber design system (Midnight #060b13, Gold #d4af37, Cyan #00e5ff), 4 720p 30fps web video preview players with poster frames, 0 broken links across 28+ projects, and WCAG AAA compliance.

Authoritative Citations & Federal References

- **U.S. Bureau of Labor Statistics (BLS):** *Occupational Outlook Handbook: Information Security Analysts*. Data release 2024–2026. <https://www.bls.gov/ooh/computer-and-information-technology/information-security-analysts.htm> (<https://www.bls.gov/ooh/computer-and-information-technology/information-security-analysts.htm>) (SOC 15-1212).
- **National Center for O*NET Development:** *15-1212.00 – Information Security Analysts*. O*NET OnLine. Sponsored by USDOL/ETA. <https://www.onetonline.org/link/summary/15-1212.00> (<https://www.onetonline.org/link/summary/15-1212.00>).
- **National Science Foundation (NSF):** *CyberCorps: Scholarship for Service (SFS) Program Guidelines*. Directorate for STEM Education. <https://new.nsf.gov/funding/opportunities/cybercorps-scholarship-service-sfs> (<https://new.nsf.gov/funding/opportunities/cybercorps-scholarship-service-sfs>) and <https://www.sfs.opm.gov/> (<https://www.sfs.opm.gov/>).
- **National Institute of Standards and Technology (NIST):** *Security and Privacy Controls for Information Systems and Organizations*. NIST SP 800-53, Rev. 5. <https://csrc.nist.gov/publications/detail/sp/800-53/rev-5/final> (<https://csrc.nist.gov/publications/detail/sp/800-53/rev-5/final>).
- **Microsoft Corporation & Certipoint:** *Microsoft Certified Educator (MCE) Digital Badge Verification*. Credly Digital Credential Platform. https://www.credly.com/badges/d4e5c326-c255-405c-b50e-0a369d6fc3a0/public_url (https://www.credly.com/badges/d4e5c326-c255-405c-b50e-0a369d6fc3a0/public_url).
- **Future Business Leaders of America (FBLA):** *2024–25 Competitive Events Guidelines & Rating Sheets: Electronic Career Portfolio*. High School Division. <https://www.fbla.org/> (<https://www.fbla.org/>).



Andrew Strachan

ELECTRONIC CAREER PORTFOLIO

Official Electronic Career Portfolio created for the Future Business Leaders of America (FBLA) National Competitive Events Program.

Institutional Contact: strachan@uab.edu · Birmingham, Alabama

PORTFOLIO SECTIONS

Interactive Resume

Career Research (SOC 15-1212.00)

Career-Related Education

Educational Enhancement

Special Skills & MCE Credential

28+ Projects & Media Showcase

Sources & Provenance Ledger

AUTHORITATIVE VERIFICATIONS

Credly MCE Badge Verification (https://www.credly.com/badges/d4e5c326-c255-405c-b50e-0a369d6fc3a0/public_url)

NSF CyberCorps SFS Program (<https://new.nsf.gov/funding/opportunities/cybercorps-scholarship-service-sfs>)

BLS Occupational Outlook Handbook (<https://www.bls.gov/ooh/computer-and-information-technology/information-security-analysts.htm>)

GitHub: @astrachan163 (<https://github.com/astrachan163>)

FBLA Rating Sheet & Guidelines (PDF)

© 2026 Andrew Strachan · NSF CyberCorps: Scholarship for Service (SFS) Fellow · All Rights Reserved.

Built for FBLA National Leadership Conference & Federal Public Service Career Pathways.